

PROFILE

Research scientist with 9+ years at the intersection of quantum many-body physics, machine learning, and high-performance computing. I develop scalable algorithms –classical and quantum– for simulating electronic matter and quantum dynamics, and I turn research into quality code. Experience spans leading cross-disciplinary teams, industry collaborations, and grant-funded programs. Recently at ETH Zürich and EPFL, I advanced neural-quantum-state methods and time-evolution schemes, with results published in Nature Communications and Nature Physics.

Focus areas: Monte Carlo methods, deep learning solutions for quantum matter, quantum device simulation (digital/analog), scientific ML for materials & condensed matter, multi-GPU acceleration.

Highlights: CNRS AI Rising Talent (€1M, 2025) • Microsoft Research Grant (CHF 190k, 2021) • Publications in Nature Communications (2024), Nature Physics (2025), NeurIPS (2023) • Co-founder of ML startup (acquired by Silverfin).

EXPERIENCE

Senior research assistant (Oberassistent), **ETH Zürich, Switzerland**

10/2024 – Present

- Developing computational methods for strongly correlated materials and quantum device simulation (digital & analog).
- Building high-performance code for large-scale many-body simulations across multi-GPU platforms.

Post-doctoral researcher, **EPFL, Switzerland**

10/2021 – 9/2024

- Designed novel time-propagator expansions & variational time-evolution methods for electronic matter.
- Built neural network architectures for quantum systems; advanced NQS for real-time dynamics.
- Secured partnership funding with Microsoft Research (Swiss JRC, CHF 190k) and delivered ML techniques for 2D materials.
- Recognized with EPFL performance bonus.
- Publications in Nature Communications (2024) and Nature Physics (2025) on quantum dynamics and simulators

University researcher, **IMEC and University of Antwerp, Belgium**

10/2020 – 9/2021

- Promoted to Principal Investigator (April 2021) to lead the AI4Science team
- Led research projects on scientific machine learning, topological data analysis, and causal machine learning
- Contributed to securing EU (H2020) funding through cross-institutional collaboration; subsequently supervised a PhD student funded under this grant.

Post-doctoral research fellow, **Ghent University, Belgium**

10/2019 – 9/2020

- Developed machine-learning models for quantum and classical physics problems, bridging statistical physics and ML.
- Designed early prototypes of neural-quantum-state approaches that later informed EPFL/ETH work.
- Supervised graduate students and provided mentorship in ML methods applied to physics.

Scientific advisor (20%), **BioClue, Belgium**

10/2019 – 9/2020

- Prototyped a novel DNA sequencing and search-and-retrieval model based on patented HYFT patterns, by building models at the intersection of biotech discovery, biotherapeutics, and artificial intelligence, contributing to €2M external investment, and acquisition by IPA (\$HYFT).

Head of machine learning & co-founder, **Boltzmann, Belgium**

10/2018 – 9/2019

- Led a 10-person ML consulting team; delivered client R&D projects and launched an NLP anomaly-detection product adopted across the Belgian bookkeeping industry.
- Company acquired by Silverfin
- Successfully solved problems in research and development for clients, application for national subsidies

SKILLS

Programming & Tools: Python, Jax, PyTorch, multi-GPU, C++, open-source software development

Algorithms & Methods: Scientific Machine Learning, Monte Carlo, quantum algorithms & tensor networks

Communication & Leadership: Expert in bridging theoretical physics, computer science, and experimental teams to align goals and methodologies; mentored 20+ graduate researchers

Additional certificates: Quantum Machine Learning (edX, 2019), Deep Learning Specialization (Coursera, 2018), Sequence Models (Coursera 2018), Convolutional Neural Networks (Coursera, 2018), Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization (Coursera, 2018), Neural Networks and Deep Learning (Coursera 2018), Structuring Machine Learning Projects (Coursera, 2018)

Outreach & Community: Lecturer, Vlerick Business School (bootcamp ML in Finance); Guest lecturer, "Big Data Science," Ghent University (UGent); Author, online course: Quantum Computing & Quantum Machine Learning (bluecourses.com), Open-source contributor.

Domains: Strongly correlated electrons & spins, out-of-equilibrium dynamics, analog & digital quantum simulators

Languages: Dutch (native), English (fluent), French (advanced), German (beginner)

EDUCATION

Doctor in Science: Physics, *Ghent University, Belgium*

10/2014 – 09/2018

- Theoretical high-energy physics, thesis: “Analyticity constraints for hadron amplitudes”
- Theoretical predictions for non-perturbative QCD experiments
- Commendations by the jury for exceptional research skills

MSc in Artificial Intelligence, *KU Leuven, Belgium*

10/2015 – 09/2020

- Courses on theory and practical concepts in computer vision, kernel methods, deep learning, information retrieval and search engines, theory of machine learning, etc
- Thesis on “Quantum machine learning”, summa cum laude, part-time study

MSc and BSc in Physics and Astronomy, *Ghent University, Belgium*

10/2009 – 07/2014

- Summa cum laude, ranked first in class

SELECTED AWARDS & FUNDING

- CNRS Choose France “**AI Rising Talent**” (2025, €1M)
- Research grant by the Microsoft Swiss Joint Research Centre (‘Swiss JRC’) (2021, CHF 190,000)
- Research and Innovation action (RIA) grant (H2020-FETOPEN) (2020, €442,809 IMEC share)
- Best student graduating in Physics – Ghent University (2014)

SELECTED PUBLICATIONS

- **Ab-initio variational wave functions for the time-dependent many-electron Schrödinger equation**, Nature Communications (2024)
First demonstration that neural networks can accurately capture electronic quantum dynamics via a new computational framework (with G. Pescia, A. Sinibaldi, G. Carleo).
- **Predicting topological entanglement entropy in a Rydberg analogue simulator**, Nature Physics (2025)
Accurate simulation framework for modern Rydberg simulators revealing properties of prepared topological states (with L. Mauron, Z. Denis, G. Carleo).
- **Wasserstein Quantum Monte Carlo**, NeurIPS (2023)
Variational framework outperforming standard VMC via imaginary-time evolution (with K. Neklyudov, L. Thiede, J. Carrasquilla, Q. Liu, M. Welling, A. Makhzani).
- **Local fermion-to-qudit mappings, arXiv:2412.05616 (2024)**
Encodes electronic DOFs onto qudits (d=4) enabling scalable parity-preserving mappings; includes circuits and Trotter decompositions (with R. Carobene, S. Barison, A. Giachero).

Full list on Google Scholar.